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Review Article

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Challenges of the allosteric regulation of hemoglobin by interaction with Cys93 of the β chain for the treatment of sickle cell disease

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ABSTRACT

Voxelotor exerts its regulatory effect on the affinity of hemoglobin for oxygen through an allosteric mechanism involving binding of the drug to the *α*-chain of hemoglobin. Cystein 93 of the *β*-chain of hemoglobin is an additional allosteric regulatory site of Hb affinity for oxygen that has been investigated for several decades as a potential target for the treatment of sickle cell disease. This review covers research, in vitro and in preclinical models, on several thiol-reactive molecules such as iodoacetamide, N-ethylmaleimide, 1-bromoacetyl-3,3-dinitroazetidine, isothiocyanate derivatives, thimerosal, glutathione, triazole disulfide, and allyl disulfide, which bind to *β*-Cysteine 93. Although these compounds were effectively employed to demonstrate that *β*-Cysteine 93 is an allosteric regulator of hemoglobin function, none of them have yet provided the safety and preclinical data to allow the initiation of clinical trials for the treatment of SCD. Such an opportunity may be provided by repositioning dalcetrapib, an orally available and safe thiol derivative currently in Phase 3 clinical trials for the prevention of cardiovascular events: dalcetrapib has been shown to reduce the incidence of cardiovascular events in patients with the AA genotype of the Adenylyl Cyclase 9, rs1967309 variant, which is prevalent in people of African descent. Recently, dalcetrapib was also observed to bind to Cysteine 93 of the *β*-chain of hemoglobin, leading to a reduction in hemoglobin S polymerization and alterations in hemoglobin interaction with oxygen. Dalcetrapib could therefore be a therapeutic that limits red blood cell sickling in SCD patients, thus mitigating the associated complications. Nonetheless, clinical trials on dalcetrapib in this population are currently lacking.

KEYWORDS: sickle cell disease (SCD), Cysteine, hemoglobin-S, allostery, dalcetrapib

IMPORTANCE OF *β*-CYS93 IN THE ALLOSTERIC MODULATION OF HEMOGLOBIN ACTIVITY

SCD is due to a mutation of Glu6 to Val6 in the β-chain of human Hb, leading to the production of an abnormal Hb, called HbS. At low oxygen pressure, HbS may polymerize, resulting in morphological (sickling) and physiological changes in RBCs. Sickled RBCs are poorly deformable and very fragile, leading SCD patients to develop vaso-occlusive-like complications and chronic hemolytic anemia.

The human Hb molecule is composed of 2 α-chains (α1 and α2) and 2 β-chains (β1 and β2), forming a heterotetramer. The binding of ligands is a cooperative process, whereby the binding of oxygen to one subunit facilitates the binding of oxygen to the other subunits. The cooperativity of Hb is associated with the allosteric transition from the T-state to the R-state (T for tense and R for relaxed), whereby the oxy or R-structure exhibits high oxygen affinity and the deoxy or T-structure exhibits low oxygen affinity.[1] The allosteric nature of Hb is a consequence of the subunits packing in the tetramer. The α1β1 and α2β2 interfaces of the dimer are highly stable, in contrast to the α1β2/α2β1 interfaces that mediate the quaternary changes associated with ligand binding. 5-Hydroxymethylfurfural (5HMF), an aromatic aldehyde, was demonstrated to interact with Hb via a reversible Schiff-base linkage to the N-terminal valine of the α-chain, thereby inducing an allosteric conformational change that increases the oxygen affinity of Hb.[2] Subsequently, Metcalf et al[3] discovered GBT-440 (voxelotor), a vanillin derivative that also binds to Val1 of the α-chain of Hb and has been the subject of extensive investigation. This discovery resulted in the development and regulatory approval of voxelotor for the treatment of SCD.[4] The identification of new allosteric regulators of hemoglobin has been investigated. It was hypothesized that some imidazolylacryloyl derivatives could bind Cys93; although the initial results were inconclusive,[5] they led to the

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synthesis of azolylacryloyl derivatives which covalently interact with βCys93[6] and display antisickling activities.[7,8] This approach has been reviewed by Safo et al[9] and more recently by the same team.[10] The continuing effort to discover more druggable compounds targeting Val1, with better pharmacokinetics, is illustrated by a recent publication.[11]

Perutz[1] previously indicated that the conformation of the reactive sulfhydryl groups of βCys93 undergoes a change when Hb changes from the R to T state. Hb contains 6 cysteine residues. The β-subunit contains 2 cysteine residues, β93Cys and β112Cys, which are associated with the 2 Hb interfaces. A cysteine is present at position β93 at the α1β2/α1β2 interface. This residue is conserved in Hb from all mammalian species,[12] which suggests an essential role for this amino acid. Including βCys93, 3 amino acids are found to be strictly conserved in the β-chain of Hb of all mammals and birds. However, only 2 of these, a histidine and a phenylalanine, are essential for stabilizing the heme moiety and facilitating oxygen transport. Deterding et al[13] identified a sequence, GTFASLSELHCDK, containing Cys93, as reactive on the β-chain of Hb. Irreversible oxidation of βCys93 ultimately leads to the collapse of the hemoglobin structure and release of heme. Human fetal hemoglobin (HbF), similarly to the adult hemoglobin (HbA), carries redox active γCys93 in the vicinity of the heme pocket. In a detailed mutagenic study of HbA and HbF, Kettisen et al[14] showed that substitution of γCys93 with Ala resulted in oxidative instability characterized by increased oxidation rates. It is noteworthy that Cys93 represents the primary glutathionylated residue of human Hb, which results in a 5–6-fold increase in the affinity of Hb for oxygen and a reduction in the cooperativity of binding.[15] The cysteine at β112 forms part of the α1β1/α2β2 interface, but exhibits less conservation among mammalian Hb than the βCys93. Hb of both hypoxia-resistant and hypoxia-unstable species are capable of glutathionylation of Cys93, which indicates the existence of a single mechanism of redox regulation of Hb function during the development of oxidative stress in hypoxia-resistant and nonresistant species.[12] Kan et al[16] employed the sulfhydryl reagent p-hydroxy-mercuri-benzoate (PMB) to demonstrate that all three cysteine residues of Hb, the surface-exposed β93Cys and the shielded α104Cys and β112Cys, could be modified. The authors reported that the modification proceeded in a stepwise manner, with the following order of reactivity: β93Cys > α104Cys > β112Cys. Interestingly, a human mutation where βCys93 is replaced by Arg was first described in 1984 in Japan and named Hb Okazaki. This rare Hb variant has high oxygen affinity but normal phenotype[17] and was also observed in Europe by Grimaldi et al[18]

Notwithstanding the fact that βCys93 is not situated in close proximity to the HbS mutation (β6Glu-Val), it has been demonstrated to be a pivotal amino acid for β subunit stability. Khan et al[19] indicated that βCys93 is situated in a conformationally plastic domain containing residues whose interactions are directly linked to the allosteric properties of the Hb tetramer. The reactivity of the βCys93 sulfhydryl group toward a range of reactants is highly sensitive to both the quaternary and tertiary conformations of Hb. It has been put forth that the in vivo functional roles of β93 are also subject to modulation by the overall protein structure.

ROLE AND IMPORTANCE OF HB CYSTEINE 93 (*β*CYS93) IN NITRIC OXIDE BIOACTIVITY AND OXIDATIVE STRESS

As previously discussed, Cys93 occupies a pivotal position at the subunit interfaces and is critically involved in the Hb R to T transition. Alayash[20] and Basu et al[21] showed that nitric oxide (NO) bound to a cysteine residue (Cys93) on the β-chain of Hb to form S-nitroso-hemoglobin (SNO-Hb). Zhang et al[22] reported that mice with a βCys93Ala mutation are deficient in hypoxic vasodilation. Their results support the hypothesis that βCys93-derived NO bioactivity mediates the autoregulation of blood flow, particularly in vascular areas where oxygen tension is low, and that βCys93 is essential for normal cardiovascular function. In the presence of a mild oxidant, βCys93 is readily and irreversibly oxidized to cysteic acid, which may therefore be used as a biomarker of oxidative stress. Alayash[20] proposed that βCys93 could be considered as an important biological marker of oxidative stability of hemoglobin within RBCs intended for transfusion or in hemoglobinopathies. The introduction of a mutation at a specific site in an amino acid with redox activity, or the implementation of direct chemical modifications that can shield βCys93, has been proposed as a potential strategy to enhance the oxidative resistance of Hb. This approach may offer a protective therapeutic strategy.

COMPOUNDS PREVIOUSLY INVESTIGATED FOR BINDING TO HEMOGLOBIN *β*CYS93

Because of the presence of reactive cysteines in a multitude of enzymes and proteins,[23] the development of compounds that react and bind to sulfur is a significant challenge. This is due to the potential formation of covalent haptens, which may trigger immune reactions, and the possibility of inefficient clearance of covalently bound thiols from plasma and tissues. As a consequence, with the exception of the marketed angiotensin-converting enzyme inhibitor captopril[24] and the phase 3 compound dalcetrapib,[25] few sulfur derivative compounds have provided the preclinical and clinical safety data allowing their successful development as drugs. Indeed, these difficulties may be reconsidered since S-S bonds are among the weakest covalent bonds, with a dissociation energy of 251 kJ/mole (60 kcal/mole), being about 5.8- and 6.9-folds weaker than C-C (348 kJ/mole) and C-H (413 kJ/mole) bonds, respectively. Although covalent, disulfide bonds are often the "weak link" in many molecules and offer the possibility to design druggable compounds binding to βCys93 and efficiently eliminated from the body.

Various thiol reactive substances

Craescu et al[26] showed that iodoacetamide or N-ethylmaleimide interacts with βCys93 in Hb, inducing NMR-detectable structural and dynamic perturbations localized at the proximal side of the heme pocket. Indeed, Garel et al[27] conducted a comparative analysis of the properties of 21 thiol reagents,

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including their binding rate to Hb in solution and within intact cells, as well as their capacity to modify the oxygen dissociation curve in intact cells. The results demonstrated that the binding of thiol reagents to Hb does occur in conditions close to physiological ones and is related to the environment of the Cys93 SH group. The majority of these compounds did not modify Hb when incubated with RBCs, most likely because of their inability to cross the cell membrane. These compounds are important chemical tools to study Hb thiol but lack selectivity and do not have a future as drugs to treat SCD. Omar et al discovered a novel class of covalent.

RRx-001

Ning[28] identified a novel compound, RRx-001 (1-bromoacetyl-3,3-dinitroazetidine), a non-explosive molecule derived from a class of solid rocket propellants. It is hypothesized that the activity of RRx-001 is associated with a nucleophilic substitution by circulating thiol compounds and covalent binding of RRx-001 to cysteinyl residues in Hb, which is then followed by the generation of nitrogen oxides. In a study by Scinciski,[29] the partitioning of total radioactivity in RBCs and plasma was determined after in vitro addition to human, rat, dog, and monkey whole blood at concentrations of 1 and 20 μM. The results indicated that, in rats, approximately 75% of the radioactivity was associated with RBCs and 25% with plasma at 30 minutes. In humans, at 30 minutes, approximately 25% of the radioactivity is associated with RBCs and 75% with plasma. Human Hb was incubated with cold RRx-001 in buffer, and a standard proteomic protocol was employed to separate and identify the tryptic peptides. The structure of the peptide GTFATLSELHCDK was confirmed by standard peptide collision-induced fragment ions, with the alkylation occurring at the βCys93. RRx-001, which demonstrated potential as a new cancer therapeutic agent in a range of cell lines and tumor models, was found to be equipotent to cisplatin, while exhibiting superior tolerability in murine models. Furthermore, RRx-001 demonstrated efficacy as a radiosensitizer in both in vitro and in vivo models.[28] RRx-001 is currently undergoing clinical trials as a novel NO and hypoxia-mediated anticancer agent, administered via intravenous injection.[30] To the best of our knowledge, RRx-001 has not been tested in the context of SCD.

Isothiocyanates

In their investigation to identify antisickling effects of isothiocyanate derivatives, which modify in a regioselective manner Hb, Park et al[31] identified 2-(N, N-dimethylamino)ethylisothiocyanate as one of the most effective compounds. The 2-(N, N-dimethylamino)ethylisothiocyanate selectively reacts with the thiol of βCys93, which, in conjunction with the cationic group, was seen to disturb the local HbS structure. The modified HbS showed an approximately 30% increase in solubility for the fully deoxygenated state, along with a significant increase in oxygen affinity. To the best of our knowledge, no data on the bioavailability and safety of such compounds are currently available.

Thimerosal

Ethylmercury thiosalicylate (thimerosal) is used as a preservative in some medicines and vaccines. Janzen et al[32] were among the first to demonstrate adducts with human and rat hemoglobin. Sales et al[33] showed that thimerosal binds to human Hb via free Cys residues, causes conformation changes and leads to harmful effects associated with O_2 transport. De Magalhaes et al[34] investigated the interaction of thimerosal with human Hb, and confirmed a stoichiometry 2:1, ligand-protein. Spectroscopic data suggested that thimerosal induces structural changes in Hb, through ethylmercury transfer to human Hb-Cys93 residues, as a consequence, inhibition of Hb-O_2 binding capacity up to 72% (human Hb), and 50% (human erythrocytes). Overall, these publications strongly support a direct interaction of thimerosal with hemoglobin, unfortunately, with toxicological consequences, preventing the chronic use of such compounds as therapeutic agents.

Glutathione

The binding of reduced glutathione (GSH) by disulfide linkage to Cys93 of Hb tetramers has been demonstrated to increase the oxygen affinity and significantly inhibit sickling of sickle RBCs at low partial oxygen pressure.[15] This phenomenon has also been observed when the sulfhydryl groups of Hb are modified by N-ethylmaleimide. Glutathione is a potent antioxidant present at mM concentrations within RBCs. It may be oxidized to form GSSG, which may participate in the process of glutathionylation of proteins via a thiol exchange with free cysteine residues of proteins. The oxygen-binding measurements revealed that GSH-Hb exhibited a nearly a 6-fold increase in oxygen affinity compared with native Hb, accompanied by minimal heme-heme interactions. In a more recent study, Das et al[35] employed molecular dynamics simulations to demonstrate that the stereo-specific localization of the glutathione moiety in the hydrophobic groove across the globin subunit interface of tetrameric HbS may contribute to the inhibition of polymerization. The authors put forth the proposition that the insights gained into the conformational changes that occur upon the inhibition of HbS polymerization by glutathionylation may be translated into the development of molecularly targeted therapeutic approaches for the treatment of SCD.

Triazole disulphide

Kassa et al[36] considered βCys93 as a potential therapeutic target in Hb and identified compounds with the potential to treat SCD. These compounds have been demonstrated to exert dual allosteric and antioxidant effects, namely, an increase in the affinity of HbS for oxygen, coupled with a prolongation of the delay time for HbS polymerisation. Nakagawa et al[37] screened 38 700 compounds using small molecule microarrays and identified 427 molecules that bind to hemoglobin and identified novel allosteric effectors for Hb and demonstrated that this allosteric effector increased the oxygen affinity of HbA without causing hemolysis. A triazole disulfide, specifically (di(5-(2,3-dihydro-1,4-benzodioxin-2-yl)-4H-1,2,4-triazol-3-yl)disulfide), TD-1 represents a prototype of such compounds, which allosterically regulate Hb through the binding of the thiol derivative TM-1. A disadvantage of TD-1 is its

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relatively poor solubility in water, which limits its use in preclinical investigations. A new derivative (4,4′-Di(1,2,3-triazolyl) disulfide, designated TD-3, was designed by Nakagawa et al[38] and proposed as a more druggable derivative of TD-1. It acts through the binding of the thiol derivative TM-3 to βCys93 and by inhibiting salt-bridge formation between β-Asp94 and β-His146, which results in a decreased magnitude of the Bohr effect and an increased affinity of HbA for oxygen. Furthermore, the researchers demonstrated that the intravenous administration of TD-3 to mice resulted in an increase in the affinity of murine Hb for oxygen. To the best of our knowledge, no further preclinical or clinical development has been published for this class of compound.

Diallyl sulphide

Garlic and aged garlic are known to contain a large amount of diallyl sulfide and are commercially available. Onishi et al[39] demonstrated that diallyl sulfide given as nutritional antioxidant supplements, hydroxyl radical scavengers, and iron-binding agents, incubated with RBCs from patients with SCD could inhibit the formation of dense cells in vitro. The concentrations at which effective nutritional supplements could inhibit dense cell formation by 50% were 4.0 mg/mL for aged garlic extract. An ex vivo study and a pilot clinical trial demonstrated that a cocktail consisting of daily doses of 6 g of aged garlic extract, 4–6 g of vitamin C, and 800-1200 IU of vitamin E may be beneficial to patients judged from the increase in hematocrite. Bonaventura et al[40] showed that garlic-derived organic disulfides, as well as pure diallyl disulfide, can modify βCys93; however, this modification of deoxyhemoglobin at the βCys93 site requires permeation of the RBC membrane.

As expected from these observations, aged garlic significantly inhibited the increase in thiobarbituric acid-reactive substances and hemolysis rate and prevented the loss of intraerythrocytic ATP and 2,3-diphosphoglycerate in oxidized erythrocytes. In addition, RBC deformability was improved.[41] Takasu et al[42] administered orally aged garlic to five patients with SCD and observed a reduction in the number of Heinz bodies over a period of 4 weeks (58.9 ± 20.0% at baseline to 29.8 ± 15.3% at follow-up, $P = .03$). Further studies are required to elucidate the physiological role of the reported garlic extract-induced allyl modification on human Hb. However, the results indicate that constituents of natural products, such as those from garlic extract and aged garlic, could contribute to the treatment of SCD, as proposed by Takasu et al[43]

DALCETRAPIB

Evidence for dalcetrapib binding to cysteine residues present in the hydrophobic pockets and regulatory sites of different proteins

Connolly et al[44] examined the effects of various cysteine-modifying reagents on the activity of human cholesteryl ester transfer protein (CETP). The transfer activity was not affected by dithiothreitol (DTT) or other alkylating agents, including iodoacetamide and N-ethyl maleimide. Nevertheless, the incubation of CETP with hydrophobic thiol-modifying reagents, such as p-chloromercuriphenylsulfonic acid, resulted in the complete, time-dependent inactivation of both the cholesteryl ester and triglyceride transfer activities. The inactivation could be prevented by the inclusion of DTT in the incubation medium. The authors put forth the hypothesis that CETP contains an indispensable free cysteine within a hydrophobic milieu of the protein.

The oral availability of the majority of sulfur-reactive compounds is extremely limited, and only a few sulfur-containing compounds have been approved and registered as therapeutic agents. In the course of developing a orally bioavailable CETP inhibitors designed to bind covalently to a cysteine residue (Cys13) located in a hydrophobic pocket of CETP, Okamoto et al[45] synthetized dalcetrapib (S-{2-([[1-(2-ethylbutyl) cyclohexyl]carbonyl] amino)phenyl}2-methylpropanethioate) a lipophilic thio-ester with a logP close to 7.0.[46] This dalcetrapib is efficiently de-esterified after oral absorption by non-specific esterases and lipases present in body fluids, such as plasma, cell homogenates, intracellular milieu or in vitro by exposure to basic pH.[47,48] The pharmacologically active free thiol form of dalcetrapib, inhibits CETP activity only when Cys13 is present. Dalcetrapib was observed to inhibit the CETP serine mutants at residues 1 and 131 (C1S and C131S), but not the C13S mutant.[49] This evidence suggests that the cysteine residue at position 13 of CETP is essential for the inhibitory activity of dalcetrapib. Although dalcetrapib may bind non-specifically to numerous free SH groups present in proteins, these interactions are weak and non-specifically bound dalcetrapib can easily be displaced by DTT. Consequently, dalcetrapib is cleared from the human body with a relatively short plasma half-life of approximately 31 hours.[46]

It is noteworthy that another protein, MD2, a regulator of the toll-like 4 receptor signaling pathway, contains a cysteine residue (Cys133) within a hydrophobic pocket. Mancek-Keber et al[49] demonstrated that dalcetrapib-thiol is capable of binding to Cys133 and that oral administration of dalcetrapib resulted in a significant inhibition of endotoxin-induced tumor necrosis factor α production in mice. These data lend support to the hypothesis that dalcetrapib interacts with cysteines through both an S-S bond and a hydrophobic interaction, facilitated by the localization of the reactive cysteine within or in close proximity to a lipophilic pocket.

The pharmacogenomic genome-wide evaluation of the Phase 3 dal-OUTCOMES study ($n = 5749$) demonstrated that dalcetrapib (600 mg per day for 31 months) decreased cardiovascular events by 39% in patients with the AA genotype of the ADCY9 rs1967309 polymorphism.[50] This genotype is present in the majority of subjects of African origin.[51] An attractive hypothesis is that SCD patients may also derive benefit from the cardiovascular effect of dalcetrapib, obviating the need for genetic pre-screening to exclude patients with the deleterious ADCY9 rs1967309 polymorphism present in a larger proportion in the Caucasian population.

βCys93 is in close proximity to the lipophilic pocket Leu88-Phe85, where the mutated Val6 of HbS binds

Himanen et al[52] and Adachi[53] have demonstrated that the side chains of Phe85 and Leu88, which are in close proximity to Cys93, become accessible to the surface during deoxygenation. Furthermore, these side chains are involved in the

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formation of the hydrophobic acceptor pocket for Val6 of HbS, which binds tightly between Phe85 and Leu88 in the hydrophobic acceptor pocket on an adjacent β-chain.

These observations indicate that βCys93 of Hb may exhibit structural similarities with Cys13 of CETP and with Cys133 of MD2. Both regulatory Cys are situated in close proximity to or within a lipophilic pocket, which may enhance hydrophobic interactions with the thiol reaction of dalcetrapib with these target molecules.

Distribution of dalcetrapib in plasma and RBC

The original drug target of dalcetrapib is CETP, which is mainly present in plasma; thus, the distribution of dalcetrapib in the RBC compartment has not been thoroughly investigated. We examined the distribution of dalcetrapib in RBCs by comparing the radioactivity in the plasma and the blood of rats, monkeys, and humans after 10 minutes incubation with ^{14}C-dalcetrapib (5.22 MBq/mg) and by calculating the blood/plasma partition ratio (Rb). The final concentrations of dalcetrapib added to the blood were 0.1, 1, and 10 mg/mL. Over this concentration range, the blood-to-plasma ratio of ^{14}C-dalcetrapib in the blood of rats, monkeys and humans was found to vary from 0.62 to 0.84[34] indicating that 16%-48% of dalcetrapib can be quantified in the RBC pool after a mere 10 minutes of incubation with blood. Following the oral administration of ^{14}C-dalcetrapib at a dose of 100 mg/kg to rats, the blood/plasma ratio at 2, 24 and 168 hours increased from 0.67 to 1.13 and 5.0, indicating a gradual distribution into RBCs and/or slow elimination from RBCs. The tissue concentrations of radioactivity indicated that the plasma half-life of radioactivity was 34.9 hours, compared to 72.5 hours in blood. This suggests that dalcetrapib is eliminated from the RBCs pool at a slower rate than from plasma.[54]

Interaction of dalcetrapib with plasma proteins compared to hemoglobin

Dalcetrapib, is a thioester that is hydrolyzed in vivo to the corresponding thiophenol, which undergoes further oxidation to form the dimer and mixed disulfides (Figure 1). For routine analysis, these forms were collectively determined as dalcetrapib-thiol following reduction under basic conditions with DTT, which breaks sulphur bonds and the free thiol is stabilized through derivatisation with NEM Quantification was made by UPLC or HPLC.[55]

A comparison of dalcetrapib recovery after dissolution in an organic solvent and incubation in PBS, plasma, PBS-albumin, PBS-hemoglobin, and total blood revealed that recovery was almost quantitative in all cases except in the presence of RBCs and purified hemoglobin.

The incubation of dalcetrapib with plasma and RBCs demonstrated that dalcetrapib binds rapidly and strongly to various RBC components, including the cell membrane, hemoglobin, and glutathione. Previous studies have demonstrated the binding of dalcetrapib to another abundant (mM) RBC component, glutathione (GSH); however, this binding is fully reversible and DTT sensitive. This evidence excludes GSH as a major component of the strong binding of dalcetrapib to RBC.

The binding to RBCs differs from the binding to plasma components such as albumin, which can be readily dissociated with the reducing agent DTT. Kuhlmann et al[56] confirmed that the extraction procedure of dalcetrapib-thiol from human plasma yielded 100% recovery. The use of a stronger disulfide reducing agent, tris(2-carboxyethyl)phosphine (TCEP), previously employed to partially dissociate labeled dalcetrapib from CETP in competitive binding studies, was also found to be ineffective. The use of a mild detergent such as Tween was also unsuccessful in dissociating dalcetrapib from its RBC binding site. Indeed, dalcetrapib exhibits a dose-dependent and saturable binding profile to purified human Hb. Saturation occurs at a molar ratio of approximately 1:2 (Hb: dalcetrapib). This is consistent with the hypothesis that one or two dalcetrapib molecules can bind to each Hb molecule. The digestion of Hb with a proteolytic enzyme allows the recovery of some of the bound dalcetrapib, although this process is not yet complete or quantitative.

The incubation of dalcetrapib (10 μM, 30 μM, and 100 μM) dissolved in PBS with RBCs for 1 and 4 hours, with a decreasing proportion of plasma, resulted in a time- and plasma concentration-dependent increase in recoverable dalcetrapib. This indicates that plasma may act as a buffer, influencing the uptake of dalcetrapib by RBCs. However, this uptake is poorly reversible, which supports the observation that the half-life of dalcetrapib in total blood is approximately twice as long as its half-life in plasma. A dose-response study was conducted to assess the extent of dalcetrapib binding to Hb. The incubation of dalcetrapib with plasma and Hb was conducted for a period of 30 minutes prior to derivatisation. Dalcetrapib demonstrated a dose-dependent binding to Hb, which was not extractable. In contrast, complete and linear recovery of dalcetrapib was observed in plasma. The incubation of dalcetrapib with purified human Hb resulted in a decrease in the Soret band (OD at 407 nm) at concentrations between 1 and 100 μM, a phenomenon that was also observed with DTT.[57] A comparable effect has been documented for other thiol-reactive substances, namely NEM and DTT, which have been demonstrated to bind to Cys93.[37]

DISCUSSION

Cysteine is frequently observed in functional sites of proteins and enzymes, but is markedly distinct from other common amino acids in proteins, exhibiting one of the lowest levels of abundance. This residue is characterized by a high degree of reactivity, polarizability, redox activity, and a strong affinity for metals, rendering it particularly responsive to the surrounding environment.[58] The S-S bond is the weakest covalent bond upon binding of cysteine to other thiol-containing molecules, and is susceptible to cleavage by changes in pH.

A multitude of compounds devoid of therapeutic utility have been demonstrated to bind to βCys93. In general, thiol reactive compounds are considered as impossible to be amenable to drug development, especially when oral and chronic administration are considered. Frequently, these compounds exhibit poor oral bioavailability, lack specificity for hemoglobin, are toxic or may raise long-term safety concerns. Even preclinical in vitro investigations are rendered difficult because of their chemical instability, tendency to form dimers.

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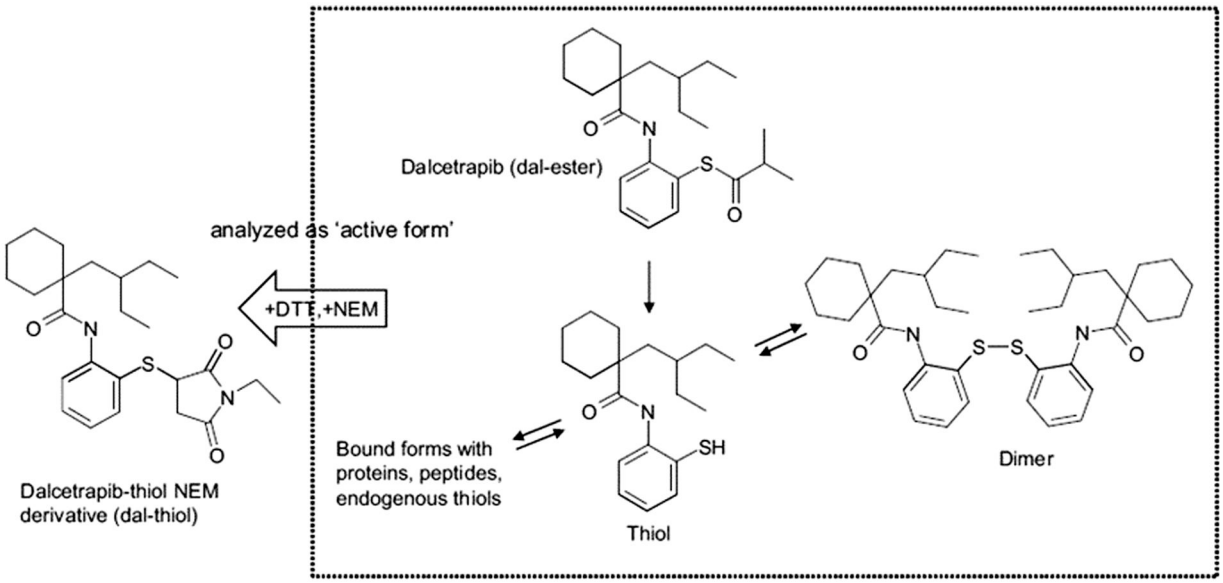


Figure 1. Dalcetrapib, a thiol ester, is rapidly de-esterified by biological fluids after intestinal absorption. The released free thiol may bind to proteins and form dimers, both of which are released in the presence of Dithiothreitol (DTT) and can be derivatized with N-Ethylmaleimide (NEM).

However, these compounds were effectively employed as tools to substantiate the assertion that βCys93 is an allosteric regulator of Hb function and a potential target for the treatment of SCD.[38] The derivative (4,4′-Di(1,2,3-triazolyl) disulfide, designated TD-3) was observed to diminish the magnitude of the Bohr effect and augment the affinity of hemoglobin for oxygen, both in vitro and in vivo, following intravenous administration of TD-3 to mice.

Two classes of compounds could be considered as potential safe and oral therapeutic agents. The first comprises allyl sulfite derivatives derived from garlic and aged garlic. It is regrettable that no definitive and convincing clinical study is currently available to the scientific community.

The second class, the thiol-reactive CETP inhibitor dalcetrapib, has been demonstrated to prevent HbS polymerization,[57] to bind strongly to hemoglobin and to enhance deformability of RBCs from SCD patients. It is of clinical importance to note that the effect of dalcetrapib on deformability was observed in RBCs from patients treated or not with hydroxyurea. Additionally, dalcetrapib has the potential to confer benefits to SCD patients by increasing HDL and lipophilic antioxidants transported by HDL, both of which are deficient in SCD patients.[59] Although dalcetrapib has been administered to thousands of patients with cardiovascular disease for several years, data on the safety and efficacy of this compound in SCD patients are not yet available if considered as a potential new treatment for SCD. The efficient partition in RBCs and strong binding of dalcetrapib to hemoglobin observed in vitro should be confirmed in vivo and demonstrated to be associated with an amelioration of RBCs rheology measured in vitro.[60]

Additional Information: Coauthor Anne Perez died.

AUTHOR CONTRIBUTIONS

Eric Joseph Niesor (Conceptualization, Methodology, Resources, Writing—original draft), Anne Perez (Data curation, Methodology), Claire Bordat (Data curation, Methodology), Fouzia Laghrissi-Thode (Funding acquisition, Resources), Elie Nader (Methodology, Supervision, Validation), and Philippe Connes (Conceptualization, Supervision, Writing—review & editing)

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None declared.

CONFLICT OF INTEREST

Hartis SA has applied for patents covering the use of Dalcetrapib for the treatment of SCD.

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